A Interfaces

Chat Messenger
web & desktop client
real-time, multi-user

Autonomous Bot
webhook-driven
long-term memory
scheduled heartbeat

OpenAI-compatible Clients
REST / SSE
SDKs, scripts

Cluster Worker Nodes
distributed task delegation

Notation

- process step
- decision
- terminal
- control flow
- LLM inference
- tool invocation

B LLM API Server

API Gateway (FastAPI) — OpenAI-compatible, streaming

`/v1/chat/completions` · JWT auth · session store · async document ingest · cluster control plane

ReAct Agent Loop

single reasoning–acting loop, iteration t = 1 … N

ACTION
Tool Dispatch
capability-aware segmented scheduling

- `file_reader`
- `grep`
- `rag`

parallel

- `file_edit`

serial barrier

- `grep`
- `rag`

parallel

read-only calls start mid-stream as soon as their arguments are parsed

Context Manager
micro-compaction of old iterations;
auto-compact on context overflow

OBSERVATION
Result Formatting
normalize output;
spill oversize result to disk (pointer kept)

ReAct cycle

request

Context Assembly
cached system prompt + workspace, env, RAG collections

THOUGHT
LLM Reasoning
streamed decoding;
tool-call parsing

Tool call(s)? — Yes → ACTION

Tool call(s)? — No → **Done-gate?**

Done-gate? — Yes → **Final Answer**

Done-gate? — No → re-enter loop

No: open to-dos remain → re-enter loop (goal mode)

Loop safeguards
iteration budget · stop signal
repeat-call / failure detection

prompt + tool schemas ↓
token stream ↑

C Tool Library

15 tools via JSON-schema function calling

Knowledge Retrieval

- ● `rag`
- ● `websearch`

hybrid search: FAISS (dense) + BM25 → RRF fusion → cross-encoder re-ranking

document QA · literature review · grounded answers

Code & Computation

- ■ `code_exec`
- ■ `shell_exec`
- ● `shell_lint`
- ■ `process_monitor`

data analysis · simulation · plotting · automation

File System & Software Engineering

- ● `file_reader`
- ● `file_navigator`
- ● `grep`
- ■ `file_edit`
- ■ `apply_patch`
- ■ `file_writer`

codebase understanding · autonomous editing · reports

Planning & Memory

- ■ `todo_write`
- ■ `memo`

task checklist · persistent key–value memory

multi-step goal execution · cross-session recall

Delegation

- ● `agent`

spawns a sub-agent (read-only explore or general) running a nested ReAct loop in a fresh context

task decomposition · parallel investigation

● concurrency-safe (parallel) ■ mutating (serial barrier)

D Local Inference Layer (on-premise)

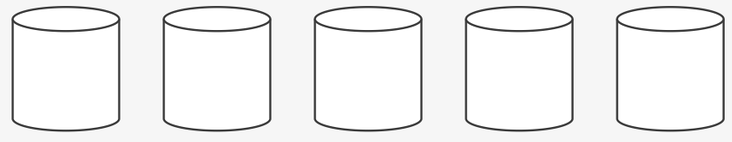
Inference Engine — vLLM	OpenAI-compatible serving · continuous batching · prefix (KV) caching · tool-call parser · reasoning parser
Open-weight LLM (swappable)	e.g., GLM · Qwen · Llama · Mistral · DeepSeek
On-premise GPU Server(s)	single node, or master + N worker nodes (cluster)

Deployment properties

- ✓ air-gapped operation
- ✓ data never leaves premises
- ✓ no per-token API cost
- ✓ model-agnostic backend
- ✓ horizontal scale-out

The agent requires only an OpenAI-compatible local endpoint: replacing the model changes no agent or tool code.

E Persistent State



Sessions — conversation history
Vector — FAISS + BM25 index
Memory — memo store
Spill — offloaded tool output
Jobs & Logs — async ingest, audit trail

tools read / write state; sessions resume across restarts